20260929 Quiz

for (MATH230) Probability & Statistics

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Exercise 4.1. The probability distribution of X, the number of imperfections per 10 meters of a synthetic fabric in continuous rolls of uniform width, is given by

x	0	1	2	3	4
$p_X(x)$	0.41	0.37	0.16	0.05	0.01

Find the average number of imperfections per 10 meters of this fabric.

Solution. By the definition of expectation,

$$\begin{aligned}\mathbb{E}[X] &= \sum_x x p_X(x) \\ &= (0)(0.41) + (1)(0.37) + (2)(0.16) + (3)(0.05) + (4)(0.01) \\ &= \boxed{0.88}.\end{aligned}$$

Thus the average number of imperfections is 0.88 per 10 meters of fabric.

□

Exercise 4.2. The probability distribution of the discrete random variable X is

$$p_X(x) = \binom{5}{x}\left(\frac{1}{4}\right)^x\left(\frac{3}{4}\right)^{5-x}, \qquad x = 0, 1, 2, 3, 4, 5.$$

Find the mean of X.

Solution. Using the given probability distribution,

$$\begin{aligned}\mathbb{E}[X] &= \sum_{x=0}^{5} x\binom{5}{x}\left(\frac{1}{4}\right)^x\left(\frac{3}{4}\right)^{5-x} \\ &= \frac{405 + 540 + 270 + 60 + 5}{1024} \\ &= \boxed{\frac{5}{4}}.\end{aligned}$$

Hence $\mu_X = 5/4$.

□

Exercise 4.5. In a gambling game, a woman is paid \$2 if she draws a jack or a queen and \$4 if she draws a king or an ace from an ordinary deck of 52 playing cards. If she draws any other card, she loses. How much should she pay to play if the game is fair?

Solution. Let X denote the amount paid to the player, excluding the entry fee. There are 8 jacks or queens and 8 kings or aces. Thus

$$\mathbb{E}[X] = 2\left(\frac{8}{52}\right) + 4\left(\frac{8}{52}\right) = \frac{12}{13} \approx 0.92.$$

For a fair game, the entry fee must equal the expected payout. Therefore, she should pay

$$\boxed{\$0.92} \quad \text{(approximately)}.$$

□

Exercise 4.7. By investing in a particular stock, a person can make a profit in one year of \$4000 with probability 0.3 or take a loss of \$1000 with probability 0.7. What is this person's expected gain?

Solution. Let X denote the gain, so a loss is represented by a negative value. Then

$$\mathbb{E}[X] = (4000)(0.3) + (-1000)(0.7) = \boxed{\$500}.$$

□

Exercise 4.9. A private pilot wishes to insure an airplane for \$200,000. The insurance company estimates that a total loss will occur with probability 0.001, a 50% loss with probability 0.01, and a 25% loss with probability 0.2. Ignoring all other partial losses, what premium should the insurance company charge each year to realize an average profit of \$500?

Solution. Let L be the insurer's loss payment. Its expected value is

$$\begin{aligned}\mathbb{E}[L] &= (200000)(0.001) + (100000)(0.01) + (50000)(0.2) \\ &= 11200.\end{aligned}$$

If p is the annual premium, an average profit of \$500 requires

$$p - 11200 = 500.$$

Hence

$$\boxed{p = \$11{,}700}.$$

□

Exercise 4.10. Two tire-quality experts examine stacks of tires and assign a quality rating to each tire on a 3-point scale. Let X denote the rating given by expert A and Y denote the rating given by B. The following table gives the joint distribution for X and Y.

$p_{X,Y}(x,y)$	y		
x	1	2	3
1	0.10	0.05	0.02
2	0.10	0.35	0.05
3	0.03	0.10	0.20

Find μ_X and μ_Y.

Solution. From the joint distribution, the marginal probabilities are

$$p_X(1) = 0.17, \quad p_X(2) = 0.50, \quad p_X(3) = 0.33,$$
$$p_Y(1) = 0.23, \quad p_Y(2) = 0.50, \quad p_Y(3) = 0.27.$$

Therefore,

$$\mu_X = \mathbb{E}[X] = (1)(0.17) + (2)(0.50) + (3)(0.33) = 2.16,$$
$$\mu_Y = \mathbb{E}[Y] = (1)(0.23) + (2)(0.50) + (3)(0.27) = 2.04.$$

Hence

$$\boxed{\mu_X = 2.16, \qquad \mu_Y = 2.04}.$$

□

Exercise 4.13. The density of the continuous random variable X, the total number of hours in units of 100 hours that a family runs a vacuum cleaner over one year, is

$$f_X(x) = \begin{cases} x, & 0 < x < 1, \\ 2 - x, & 1 \le x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the average number of hours per year that families run their vacuum cleaners.

Solution. Since X is measured in units of 100 hours,

$$\begin{aligned} \mathbb{E}[X] &= \int_0^1 x f_X(x)\, \mathrm{d}x + \int_1^2 x f_X(x)\, \mathrm{d}x \\ &= \int_0^1 x^2\, \mathrm{d}x + \int_1^2 x(2 - x)\, \mathrm{d}x = 1. \end{aligned}$$

Thus the average running time is

$$\boxed{\text{100 hours per year}}.$$

□

Exercise 4.17. Let X be a random variable with the following probability distribution:

x	-3	6	9
$p_X(x)$	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{1}{3}$

Find $\mu_{g(X)}$, where $g(X) = (2X + 1)^2$.

Solution. By the expectation formula for a function of a discrete random variable,

$$\begin{aligned} \mu_{g(X)} = \mathbb{E}[g(X)] &= 25\left(\frac{1}{6}\right) + 169\left(\frac{1}{2}\right) + 361\left(\frac{1}{3}\right) \\ &= \boxed{209}. \end{aligned}$$

□

Exercise 4.18. Let X have the binomial distribution

$$p_X(x) = \binom{5}{x}\left(\frac{1}{4}\right)^x\left(\frac{3}{4}\right)^{5-x}, \qquad x = 0, 1, 2, 3, 4, 5.$$

Find the expected value of $g(X) = X^2$.

Solution. Using the given distribution,

$$\begin{aligned}\mathbb{E}[X^2] &= \sum_{x=0}^{5} x^2 p_X(x) \\ &= \frac{1^2(405) + 2^2(270) + 3^2(90) + 4^2(15) + 5^2(1)}{1024} \\ &= \boxed{\frac{5}{2}}.\end{aligned}$$

□

Exercise 4.19. A large industrial firm purchases several new computers at the end of each year, the exact number depending on the frequency of repairs in the previous year. Suppose the number of computers X purchased each year has the following probability distribution:

x	0	1	2	3
$p_X(x)$	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{2}{5}$	$\frac{1}{5}$

If the cost is \$1200 per unit and a refund of $50X^2$ dollars is issued at the end of the year, how much can the firm expect to spend on new computers this year?

Solution. If $g(X)$ is the net expenditure, then

$$g(X) = 1200X - 50X^2.$$

Thus

$$\begin{aligned}\mathbb{E}[g(X)] &= (0)\left(\frac{1}{10}\right) + (1150)\left(\frac{3}{10}\right) + (2200)\left(\frac{2}{5}\right) + (3150)\left(\frac{1}{5}\right) \\ &= \boxed{\$1855}.\end{aligned}$$

□

Exercise 4.20. A continuous random variable X has the density function

$$f_X(x) = \begin{cases} \dfrac{1}{2}e^{-x/2}, & x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the expected value of $g(X) = e^{X/4}$.

Solution. By the expectation formula for a function of a continuous random variable,

$$\begin{aligned}\mathbb{E}[g(X)] &= \int_0^\infty e^{x/4}\left(\frac{1}{2}e^{-x/2}\right)\,\mathrm{d}x \\ &= \frac{1}{2}\int_0^\infty e^{-x/4}\,\mathrm{d}x = \boxed{2}.\end{aligned}$$

□

Exercise 4.23. Suppose that X and Y have the following joint probability function:

$p_{X,Y}(x,y)$	x	
y	2	4
1	0.15	0.10
3	0.25	0.25
5	0.15	0.10

(a) Find the expected value of $g(X,Y) = XY^2$.

(b) Find μ_X and μ_Y.

Solution. (a) By the expectation formula for a function of two discrete random variables,

$$\begin{aligned}\mathbb{E}[XY^2] &= \sum_x \sum_y xy^2 p_{X,Y}(x,y) \\ &= (2)(1)^2(0.15) + (4)(1)^2(0.10) + (2)(3)^2(0.25) + (4)(3)^2(0.25) \\ &\quad + (2)(5)^2(0.15) + (4)(5)^2(0.10) \\ &= \boxed{31.7}.\end{aligned}$$

(b) The marginal probabilities are

$$p_X(2) = 0.55, \qquad p_X(4) = 0.45,$$

and

$$p_Y(1) = 0.25, \qquad p_Y(3) = 0.50, \qquad p_Y(5) = 0.25.$$

Hence

$$\begin{aligned}\mu_X &= \mathbb{E}[X] = (2)(0.55) + (4)(0.45) = 2.9, \\ \mu_Y &= \mathbb{E}[Y] = (1)(0.25) + (3)(0.50) + (5)(0.25) = 3.00.\end{aligned}$$

Therefore

$$\boxed{\mu_X = 2.9, \qquad \mu_Y = 3.00}.$$

□

Exercise 4.26. Let X and Y have joint density function

$$f_{X,Y}(x,y) = \begin{cases} 4xy, & 0 < x, y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the expected value of $Z = \sqrt{X^2 + Y^2}$.

Solution. Using the joint density,

$$\begin{aligned}\mathbb{E}[Z] &= \int_0^1 \int_0^1 4xy\sqrt{x^2+y^2}\, \mathrm{d}x\, \mathrm{d}y \\ &= \frac{4}{3}\int_0^1 y\left[(1+y^2)^{3/2} - y^3\right] \mathrm{d}y \\ &= \frac{4}{3}\left(\frac{1}{5}\left[(1+y^2)^{5/2}\right]_0^1 - \left[\frac{y^5}{5}\right]_0^1\right) \\ &= \frac{4}{3}\left(\frac{4\sqrt{2}-1}{5} - \frac{1}{5}\right) \\ &= \boxed{\frac{16\sqrt{2}-8}{15}}.\end{aligned}$$

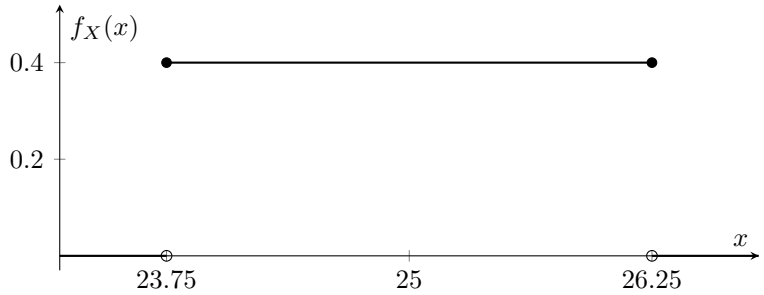
□

Exercise 4.28. The weight X (ounces) of cereal in a box has density

$$f_X(x) = \begin{cases} \dfrac{2}{5}, & 23.75 \le x \le 26.25, \\ 0, & \text{elsewhere.} \end{cases}$$

(a) Plot the density function.
(b) Compute the expected value, or mean weight, in ounces.
(c) Are you surprised at your answer in (b)? Explain why or why not.

Solution. (a) The density is constant on $[23.75, 26.25]$ and zero elsewhere.



(b) By the definition of expectation,

$$\begin{aligned} \mathbb{E}[X] &= \int_{23.75}^{26.25} x \left(\frac{2}{5}\right) \mathrm{d}x \\ &= \frac{1}{5}\left(26.25^2 - 23.75^2\right) \\ &= \boxed{25 \text{ ounces}}. \end{aligned}$$

(c) No. The density is constant on an interval symmetric about 25, so its mean is the midpoint of the interval, 25 ounces.

□